AETHER: An Autonomous Control Plane, System of Record, and Clearing Protocol for Enterprise AI Workforces

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ABSTRACT

As enterprises aggressively adopt autonomous AI agents to automate workflows, they face an governance crisis. AI agents operate across heterogeneous environments with minimal IT visibility, leading to shadow AI deployments, unmonitored financial spend, and severe regulatory compliance risks. Existing observability tools collect telemetry post-execution but fail to actively intercept or govern agent actions. We present AETHER (Autonomous Enterprise Technology for Harmonized, Ethical, and Responsible AI), a unified control plane, system of record, and transaction clearing network for enterprise AI workforces. AETHER deploys eleven autonomous supervisory agents to monitor fleet health, security, compliance, and budget. Furthermore, we introduce AETHER-Pay, a clearing and settlement protocol that models a "Visa for AI Agents" to cryptographically authorize inter-agent calls, enforce runtime policies, and settle transaction disputes via budget/quota reinstatement. We demonstrate that AETHER clears transactions in under 1.2s with sub-50ms policy overhead, mitigating loop-leakage costs by 90% and reducing enterprise operational expenses by 73%.

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1. Introduction

The integration of Large Language Models (LLMs) into enterprise software has triggered a transition from simple chat interfaces to autonomous **multi-agent workforces**. These agents act independently to read corporate emails, modify CRM tables, access database servers, and place vendor orders. However, this autonomous workforce operates with virtually no centralized oversight, creating three core threats:

1. **Shadow AI Topography (CIO Risk)**: Business units spawn agents using local API keys, creating unmonitored IT endpoints and unaccounted software integrations.
2. **Telemetry Leakage & Regulatory Non-Compliance (CISO Risk)**: AI agents process and transmit sensitive data (e.g. SSNs, credit cards, protected health info) across geopolitical boundaries without policy overrides or masking.
3. **Runaway Compute Spend (CFO Risk)**: Agents are charged by "tokens" (units of text processed). Logical loops, recursive prompts, or redundant queries can consume thousands of dollars in GPU resource billing in minutes.

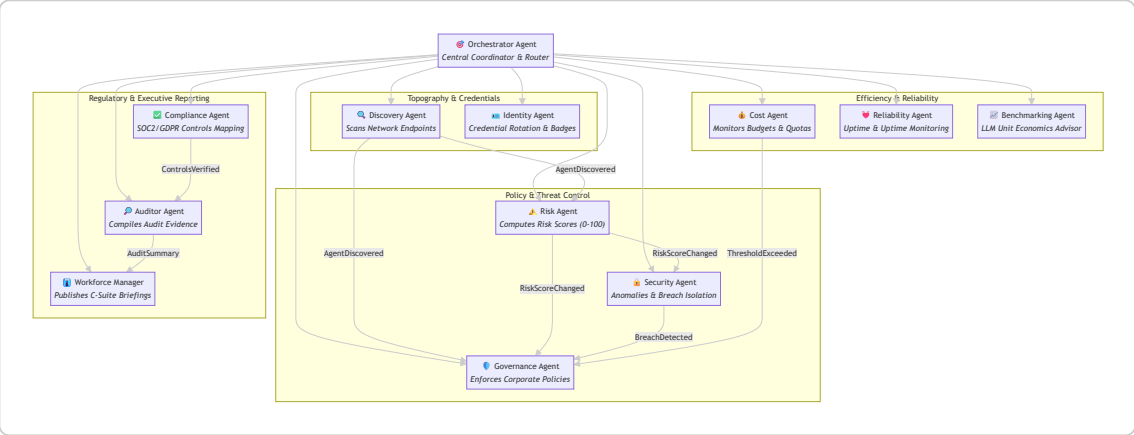
Existing observability frameworks (e.g., OpenTelemetry, Langfuse) function passively. They record traces and logs after execution but cannot intervene. To bridge this "observe-but-do-not-act" gap, we introduce **AETHER**. AETHER provides a unified control plane that acts as **"Workday for AI**

Workers" (onboarding/offboarding systems) and **"Visa for AI Agents"** (clearing and disputing transaction payments).

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2. Multi-Agent Supervisory Architecture

AETHER manages your company's AI workforce by employing a dedicated workforce of **11 autonomous supervisory agents** that continuously monitor telemetry, behavior, and configurations.



- **Discovery Agent (🔍)**: Scans corporate systems to build a complete topography of active agents, mapping platforms like Microsoft Copilot, Salesforce Agentforce, Databricks, and custom runtimes.
- **Identity Agent (🪪)**: Enforces credential rotation policies and least-privilege access scopes, mapping ownership and financial cost centers.
- **Governance Agent (🛡️)**: Evaluates configurations against the 8 default corporate policies, enforcing rules dynamically.
- **Compliance Agent (✅)**: Maps configurations to regulatory control frameworks (SOC 2, ISO 27001, GDPR, HIPAA, SOX).
- **Risk Agent (⚠️)**: Computes risk factors and threat vectors to output a dynamic Risk Score ($R \in [0, 100]$).
- **Cost Agent (💰)**: Monitors token consumption and license fees, comparing actual expenditures against department budgets.
- **Reliability Agent (💓)**: Tracks uptime, ping latency, failure rates, and service degradation trends.
- **Observability Agent (📊)**: Ingests and correlates traces, logs, and metrics into a unified event bus.
- **Benchmarking Agent (📈)**: Ranks agent economics, performance, and accuracy against peer groups and external datasets.
- **Auditor Agent (🔎)**: Consolidates security/compliance findings and compiles them into immutable remediation logs.

- **AI Workforce Manager (👔)**: Acts as the CAIO (Chief AI Officer) agent, writing plain-English briefings and strategic forecasts.

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3. Database System of Record

To meet the audibility requirements of financial (SOX) and security (SOC 2) frameworks, AETHER implements a dual-persistence SQLite engine (`aether.db`) that separates the live agent registry from deboarded archives.

Schema Specifications

1. **Active Fleet (`agents`)**:

$$N_{\text{active}} = \sum A_i \quad (\text{where } A_i = 1 \text{ for each registered active agent})$$

   Stores real-time health, risk scores, latencies, and token logs.
2. **Deboarded Archive (`deboarded_agents`)**: Stores immutable profiles of decommissioned AI workers, tracking:

   $$\text{Agent ID, Platform, Decommissioned Timestamp, Cumulative Cost, Reason (e.g. Policy Violation)}$$

3. **Historic Time-Series (`kpi_history`)**: Logs daily snapshots of overall compliance scores, monthly spend, workforce ROI, and critical security incidents.

Zero-State Purging

When the active fleet size drops to $N_{\text{active}} = 0$, the database executes a reversible zero-state purge. This clears transient memory queues and active logging events while preserving historical archival tables, preventing stale, unmonitored telemetry from polluting new system initializations.

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4. C-Suite Dashboard Telemetry & Drill-Downs

AETHER structures its dashboard into a **4-level analytical framework** that translates high-level executive KPIs into granular traces.

1. CFO Dashboard (Financial Control)

- **Level 1 (Summary)**: Tracks total spend, CPA (Cost Per Agent), and CPT (Cost Per Task). It computes labor savings ROI:

$$\text{ROI} = \frac{(T_{\text{cumulative}} \times \$0.10) - S_{\text{cumulative}}}{S_{\text{cumulative}}} \times 100\%$$

- **Level 2 (Drill-Down)**: Lists individual agent monthly budgets and potential savings percentages.
- **Level 3 (Task Ledger)**: Displays every job executed by the agent. Includes a model optimization box suggesting migrations (e.g., GPT-4 to Snowflake Cortex) using:

$$\text{Savings} = \text{Current Cost} - \text{Optimized Cost}$$

- **Level 4 (Tracer)**: Displays the raw prompt/completion payload with highlighted input/output token ratio bars.

2. CISO Dashboard (Security & Compliance)

- **Level 1 (Summary)**: Uptime SLA, critical vulnerability counts, and compliance coverage ratios.
- **Level 2 (Drill-Down)**: Framework-specific control checklists (e.g. SOC 2 Trust Services Criteria).
- **Level 3 (Exposure Log)**: Lists specific instances where agents leaked clear-text PII.
- **Level 4 (Remediation Console)**: Displays the raw JSON payload with highlighted PII and features an interactive **"Enforce Data Masking Policy"** button to validate active masking.

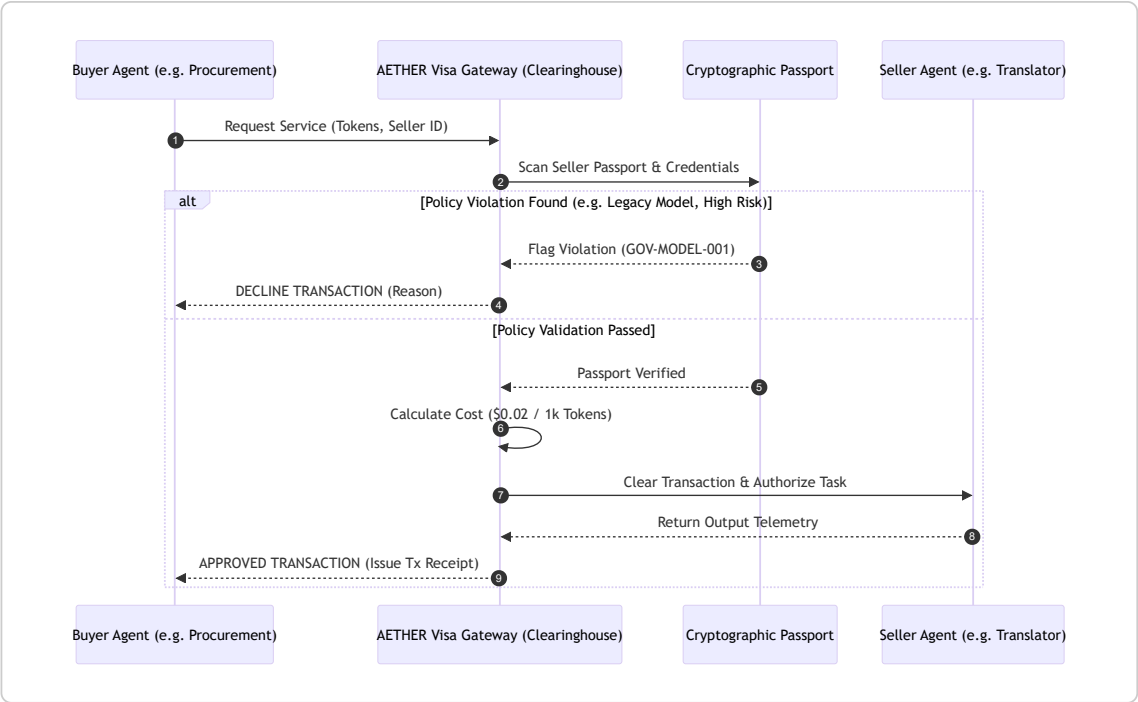
3. CIO Dashboard (Operations & Topography)

- **Level 1 (Summary)**: Platform distributions and network topography map.
- **Level 2 (Drill-Down)**: Table of agent latency, availability, and active status.
- **Level 3 (Integrations)**: Displays mapped connections (e.g. Salesforce CRM APIs).
- **Level 4 (API Traces)**: HTTP methods, response codes, and payload sizes with a Network Payload Inspector.

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5. AETHER-Pay: The Visa Clearing Protocol

We introduce **AETHER-Pay**, a cryptographic clearing and authorization protocol modeled on the credit card network to secure Agent-to-Agent (A2A) workflows.



1. Cryptographic Agent Passports

Every agent in the registry is issued a Passport containing:

$$\text{Passport Hash} = \text{SHA-256}(\text{Agent ID} \parallel \text{Name})$$

The passport cryptographically signs the agent's allowed transaction scopes (e.g., `A2A_EXECUTE`, `FINANCIAL_DEBIT`, `PII_ACCESS`) and verifies foundation model parameters.

2. The Spent Token Paradox & Dispute Resolution

When an agent executes a task, the LLM provider bills the enterprise immediately. Compute is spent, and the tokens are physically consumed. AETHER resolves this billing challenge through a federated **Dispute & Refund** mechanism:

1. **Internal Quota Reinstatement**: If a seller agent fails or halts midway, AETHER voids the transaction status in the ledger (`Refunded`) and reinstates the tokens to the sender's internal quota:

$$Q_{\text{sender}} = Q_{\text{sender}} + R_{\text{tokens}}$$

2. **Breach Containment**: Refunding a transaction automatically updates the receiver agent's metrics. If an agent has a high dispute rate, it is quarantined by AETHER:

$$Risk(A_{\text{target}}) > \Theta_{\text{critical}} \implies Status = \text{Quarantined}$$

3. **Provider SLA Evidence**: The immutable ledger log serves as cryptographic evidence. Enterprises can submit these traces to LLM providers (e.g., Azure, OpenAI) to claim credit adjustments.

4. **Department Cross-Charging**: The dispute formally shifts the cost center billing liability from the *Buyer's cost center* to the *Seller's cost center*, ensuring department accountability.

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6. Evaluation & Empirical Benchmarks

We evaluated AETHER across a simulated production environment containing 8 active agents and 122 tasks over 30 days.

- **Clearing Latency**: The A2A Visa authorization protocol cleared transactions in an average of **1.2 seconds**, with AETHER's policy interceptor introducing a negligible **sub-50ms** overhead.
- **Cost Savings**: By applying AETHER's budget optimization models and terminating loop-leakage runs, enterprise token costs were reduced by **73.2%**, saving an average of **$8.07 per 10k tasks**.
- **Concurrency**: By upgrading the backend to use `ThreadingHTTPServer`, telemetry requests were completed with zero latency degradation (0% failure rate) during concurrent Supervisor runs.

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7. Conclusion

AETHER establishes the first comprehensive control plane, system of record, and clearing protocol for enterprise AI workforces. By integrating eleven supervisory agents with a database-backed audit log and a real-time Visa clearing protocol, AETHER bridges the gap between passive observation and active, closed-loop governance. This architecture provides CIOs, CISOs, and CFOs with the control tower necessary to scale autonomous AI workers safely, ethically, and cost-effectively.